

A
MAJOR PROJECT
ON
HIGHLY AVAILABLE WEB APPLICATION ON AWS
Submitted in partial fulfilment for the award of the degree of
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COMPUTER APPLICATION



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(April 2026)

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Abstract

This project presents the design and deployment of a highly available and secure web application on Amazon Web Services (AWS) cloud infrastructure using a three-tier architecture (Web, Application, and Database) deployed across two Availability Zones. The architecture incorporates Auto Scaling Groups for automatic instance scaling based on CPU utilization, Application Load Balancers for efficient traffic distribution, and Amazon RDS (MySQL) for managed database services, while a Virtual Private Cloud (VPC) with public and private subnets ensures network isolation and a NAT Gateway enables secure internet access for private instances. Amazon CloudFront provides HTTPS support with SSL certificates, and Amazon S3 is used to host static frontend files. Security is enforced through JWT-based authentication, OTP verification via Amazon SES, rate limiting, and a three-attempt login lock mechanism with SNS notifications for security alerts, and the entire infrastructure is deployed within AWS Free Tier limits, demonstrating that production-ready, highly available, and secure cloud applications can be built in a cost-effective manner.

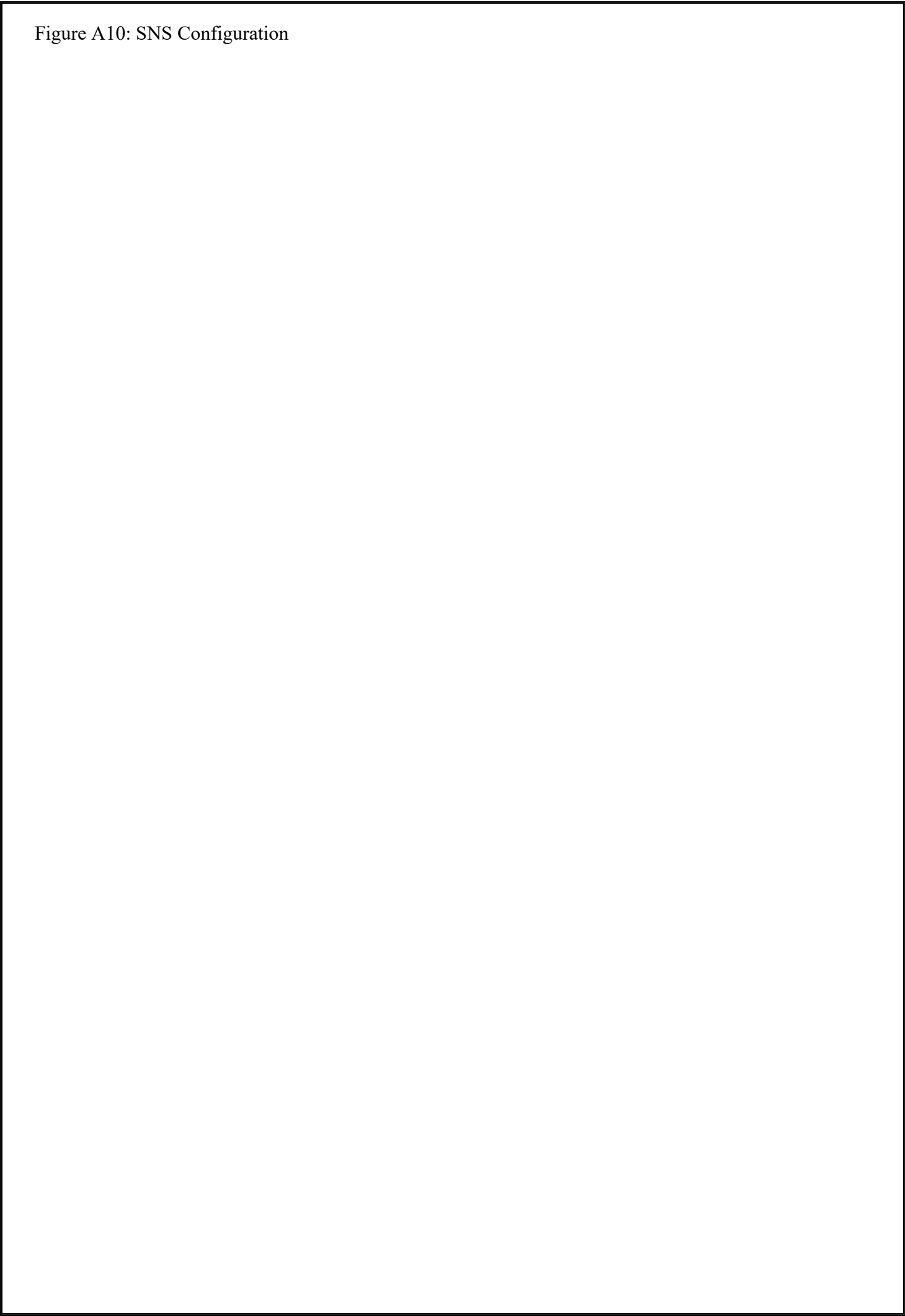
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Chapter 1

Introduction & Objectives

1.1 Introduction

Cloud computing has significantly transformed the way modern web applications are designed, deployed, and maintained. Traditional on-premise hosting models suffer from several limitations, including single points of failure, inability to handle sudden traffic spikes, dependency on manual intervention for recovery, and high maintenance costs. When an application is hosted on a single server, any hardware failure, software crash, or network issue can render the entire system unavailable, resulting in loss of users, revenue, and reputation.

Amazon Web Services (AWS) provides a comprehensive cloud platform that addresses these challenges by offering services designed for high availability, scalability, and security. High availability ensures that a system remains operational even when individual components fail. This is achieved by deploying applications across multiple Availability Zones (AZs), which are physically separate data centers within a region. In case one AZ fails, traffic is automatically redirected to healthy zones without noticeable downtime.

Scalability allows applications to automatically handle varying levels of traffic. Auto Scaling Groups (ASGs) monitor system metrics such as CPU utilization and dynamically adjust the number of running instances by launching or terminating them as required. This ensures optimal performance during high demand while minimizing resource usage during low traffic.

Security in cloud environments requires a layered approach. AWS Virtual Private Cloud (VPC) enables the creation of isolated network environments with public and private subnets. Web servers are deployed in public subnets with controlled internet access, whereas application servers and databases are placed in private subnets to prevent direct exposure. NAT Gateways allow private instances to securely access the internet for updates. Additionally, Security Groups act as virtual firewalls, permitting only authorized traffic.

This project focuses on designing and deploying a highly available and secure three-tier web application on AWS. The application, named “**Shorto – Flying Messages**”, is a real-time platform that allows users to send anonymous messages, interact with random users, send connection requests, and communicate through chat. The system is deployed across two Availability Zones and incorporates Auto Scaling, Load Balancing, managed database services, and multiple security mechanisms, all within AWS Free Tier limits.

Key Cloud Concepts Implemented

Concept	Implementation in this Project
High Availability	Web and Application servers deployed across two Availability Zones
Fault Tolerance	Auto Scaling Groups replace failed instances automatically
Scalability	Instances scale between 2 to 4 based on CPU utilization
Load Balancing	External ALB for web traffic, Internal ALB for application traffic
Network Isolation	VPC with public (Web) and private (App + Database) subnets
Security	Security Groups, JWT authentication, OTP verification
Monitoring	CloudWatch metrics and alarms
Content Delivery	CloudFront CDN with SSL/TLS
Alerting	SNS notifications for security and scaling events

1.2 Objectives

The primary objectives of this project are outlined below:

1.2.1 Cloud Infrastructure Design

Objective	Description
O1	Design a VPC with public and private subnets across two Availability Zones
O2	Deploy Web Tier (Nginx) in public subnets using Auto Scaling Group (2–4 instances)
O3	Deploy Application Tier (Node.js) in private subnets using Auto Scaling Group (2–4 instances)
O4	Deploy Database Tier (RDS MySQL) in private subnet with automated backups
O5	Configure NAT Gateways for secure internet access from private subnets
O6	Implement external and internal Application Load Balancers

1.2.2 High Availability Implementation

Objective	Description
O7	Configure Auto Scaling based on CPU utilization (>70% scale-out, <30% scale-in)
O8	Implement health checks for automatic replacement of failed instances
O9	Distribute instances across two Availability Zones for fault tolerance
O10	Configure CloudFront CDN for HTTPS and faster content delivery

1.2.3 Security Implementation

Objective	Description
O11	Implement strict Security Group rules (least privilege principle)
O12	Deploy JWT-based authentication for secure API access
O13	Implement OTP verification via AWS SES
O14	Configure rate limiting (100 requests per 15 minutes)
O15	Implement 3-attempt login lock for 24 hours
O16	Configure SNS alerts for security events

1.2.4 Application Features

Objective	Description
O17	User registration with email OTP verification
O18	User login with JWT authentication
O19	Flying messages feature (send, catch, pass)
O20	Connection request system
O21	Real-time chat using WebSocket (Socket.io)
O22	User profile management and password recovery

1.2.5 Monitoring and Cost Optimization

Objective	Description
O23	Configure CloudWatch monitoring and alerts
O24	Deploy entire system within AWS Free Tier limits
O25	Document architecture for future reference

Chapter 2

Literature Survey

2.1 Introduction

Cloud computing has significantly transformed the development and deployment of modern web applications. Amazon Web Services (AWS), being a leading cloud platform, provides a wide range of services that support high availability, scalability, load balancing, and security. This chapter reviews existing research and studies related to these core areas, forming the foundation for the proposed system.

2.2 High Availability in AWS

Varia (2010) from the AWS Architecture Center established best practices for building fault-tolerant applications. The study emphasized the use of multiple Availability Zones (AZs), Elastic Load Balancing, and Auto Scaling. The key finding was that distributing application components across multiple AZs significantly reduces the risk of system failure.

Kaur et al. (2019) conducted an analysis of five high availability patterns in AWS, including Multi-Server, Multi-Datacenter, Floating IP, Deep Health Check, and Multi-Cloud patterns. The study concluded that combining the Multi-Server pattern with Deep Health Checks provides the most cost-effective solution for small to medium-scale applications.

Kingsley (2020) demonstrated the implementation of a three-tier architecture using AWS Free Tier services. The study showed that web, application, and database tiers can be deployed across multiple Availability Zones using Auto Scaling Groups and Application Load Balancers, while remaining within free tier limitations.

2.3 Auto Scaling and Load Balancing

Angelis and Kousiouris (2021) surveyed self-adaptive cloud computing patterns and identified feedback loop structures (Monitor–Analyze–Plan–Execute) as the dominant approach in cloud automation. The study highlighted that Auto Scaling based on CPU utilization is widely adopted and found in a majority of cloud-based systems.

Research on load balancing indicates that Application Load Balancers (ALBs) provide efficient traffic distribution with minimal failover time. ALBs support advanced features such as path-based routing, WebSocket connections, and health checks, which automatically remove unhealthy instances to maintain system reliability.

2.4 Cloud Security

Chen et al. (2018) proposed a layered security framework for cloud-based applications. This framework includes network-level security (VPC and Security Groups), host-level protection, application-level security (JWT authentication), and data encryption. The study concluded that layered security provides stronger protection compared to single-layer or perimeter-based approaches. Kasmar et al. (2021) demonstrated that techniques such as rate limiting and failed login attempt tracking are effective in preventing brute-force attacks. Their research also highlighted that integrating security monitoring with auto-scaling systems ensures consistent protection during scaling events.

AWS (2022) security best practices recommend implementing the principle of least privilege using IAM roles, configuring strict security group rules, deploying sensitive resources in private subnets, and adopting a defense-in-depth strategy.

2.5 Real-Time Communication

Luo et al. (2015) evaluated the performance of WebSocket in cloud environments and found that it reduces latency by approximately 70% compared to traditional HTTP polling. This makes WebSocket highly suitable for real-time applications such as chat systems.

Prabhu and Sridharan (2019) proposed an architecture for real-time communication in auto-scaling environments. Their approach utilizes Redis as a message broker to synchronize WebSocket connections across multiple server instances, enabling scalability in real-time applications.

2.6 Summary of Literature

Author(s)	Year	Focus Area	Key Finding
Varia	2010	High Availability	Multi-AZ deployment is essential
Kaur et al.	2019	HA Patterns	Multi-Server + Health Checks is effective
Kingsley	2020	Three-Tier Architecture	Free Tier supports basic HA
Angelis & Kousiouris	2021	Auto Scaling	Feedback loop is dominant pattern
Chen et al.	2018	Security	Layered security is necessary
Kasmar et al.	2021	Security Monitoring	Rate limiting prevents brute-force attacks
Luo et al.	2015	WebSocket	70% latency reduction

Chapter 3

Research Gap & Proposed System

3.1 Research Gaps

Based on the detailed literature survey presented in Chapter 2, the following research gaps have been identified in the existing body of work:

3.1.1 Gap 1: Complete High Availability Stack on AWS Free Tier

Aspect	Existing Work	Gap
Multi-AZ Deployment	Discussed by Varia (2010) and Kingsley (2020)	✗ No complete implementation
VPC with Public/Private Subnets	Mentioned in AWS documentation	✗ Not integrated with Auto Scaling
NAT Gateway Configuration	Standard AWS practice	✗ Missing in Free Tier implementations
Dual ALB (External + Internal)	Recommended by AWS	✗ Rarely implemented together
ASG for Web and App Tiers	Discussed theoretically	✗ No practical demonstration

Specific Gap:

There is no existing study that provides a complete, end-to-end highly available architecture integrating VPC, subnets, NAT Gateway, dual load balancers, Auto Scaling Groups, and RDS within AWS Free Tier limits along with practical implementation.

3.1.2 Gap 2: Security Integration with Auto Scaling

Aspect	Existing Work	Gap
JWT Authentication	Well documented	✓ Available but not integrated
OTP Verification	Common practice	✗ No SES integration with ASG
Rate Limiting	Discussed by Kasmar et al. (2021)	✗ Not tested in scaling scenarios
Account Lock Mechanism	Standard practice	✗ No SNS integration
Security Propagation	Manual setup	✗ Not automated during scaling

Specific Gap:

The integration of security mechanisms with auto-scaling infrastructure, especially how security configurations propagate to new instances and how alerts are generated, is not well documented.

3.1.3 Gap 3: Real-Time Communication in Auto-Scaling Environments

Aspect	Existing Work	Gap
WebSocket Performance	Luo et al. (2015)	✓ Well established
Single-server WebSocket	Common practice	✓ Available
Multi-server WebSocket	Prabhu & Sridharan (2019)	⚠ Requires additional infrastructure
WebSocket with ASG	Not addressed	✗ Major gap
Connection Handover	Not studied	✗ Missing

Specific Gap:

Maintaining stable WebSocket connections during auto-scaling events remains a challenge. Existing solutions increase cost and complexity, making them unsuitable for Free Tier environments.

3.1.4 Gap 4: Automated Security Alerting using SNS

Aspect	Existing Work	Gap
SNS Alerts	Supported by AWS	✓ Available
Failed Login Alerts	Basic implementations exist	⚠ Limited integration
Account Lock Alerts	Common practice	✗ Not integrated with SNS
Scaling Alerts	Not addressed	✗ Missing
Health Monitoring Alerts	Not addressed	✗ Missing

Specific Gap:

A fully integrated SNS-based alerting system covering security and operational events is not well documented.

3.1.5 Gap 5: Cost-Effective High Availability

Aspect	Existing Work	Gap
Free Tier Limits	Documented by AWS	✓ Available
Single-AZ Database	Demonstrated	✓ Available
Multi-AZ Compute	Discussed	✓ Available
Complete Free Tier HA	Not available	✗ Major gap
Cost vs Availability	Not analyzed	✗ Missing

Specific Gap:

There is no comprehensive reference architecture that balances high availability and cost efficiency within AWS Free Tier constraints.

3.2 Summary of Research Gaps

Gap ID	Description	Severity	Impact
G1	Complete HA stack on AWS Free Tier	High	Limits adoption
G2	Security with auto-scaling	High	Security risks
G3	Real-time scaling challenges	Medium	Limits chat systems
G4	SNS alert integration	Medium	Slow response to issues
G5	Cost-effective HA design	High	No budget-friendly reference

3.3 Proposed System

3.3.1 Overview

The proposed system addresses the identified gaps by implementing a complete highly available and secure web application on AWS Free Tier. It integrates VPC, multi-AZ deployment, Auto Scaling Groups, Load Balancers, RDS database, CloudFront CDN, and advanced security mechanisms including JWT authentication, OTP verification, rate limiting, and SNS-based alerts.

3.3.2 Gap-wise Solution

Gap	Solution
G1	Full architecture with VPC, subnets, NAT Gateway, dual ALB, ASG
G2	Automated security using IAM roles and user data scripts
G3	WebSocket with sticky sessions and database-based state management
G4	CloudWatch + SNS for real-time alerts
G5	Optimized design within AWS Free Tier

3.3.3 Proposed Architecture Components

Component	Specification	Gap Addressed
VPC	10.0.0.0/16 with subnets G1	
Availability Zones	Two AZ deployment	G1
Web Tier	EC2 + Nginx + ASG	G1
App Tier	Node.js + ASG	G1
Database	RDS MySQL	G5
Load Balancer	External + Internal ALB	G1
CloudFront	CDN + HTTPS	G1
Security Groups	Least privilege rules	G2
SNS	Alert notifications	G4
SES	OTP emails	G2
CloudWatch	Monitoring	G4
S3	Storage(logs,static files)	G1
IAM	Secure Service Access	G2

Figure 1: System Architecture Diagram

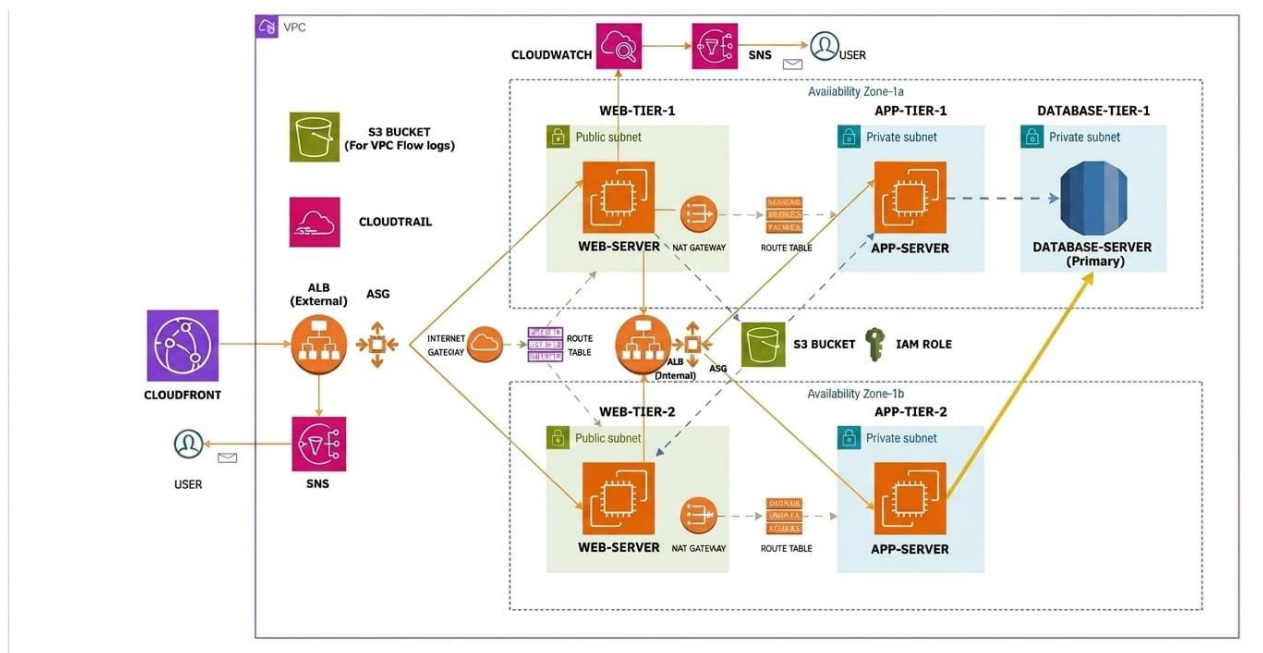
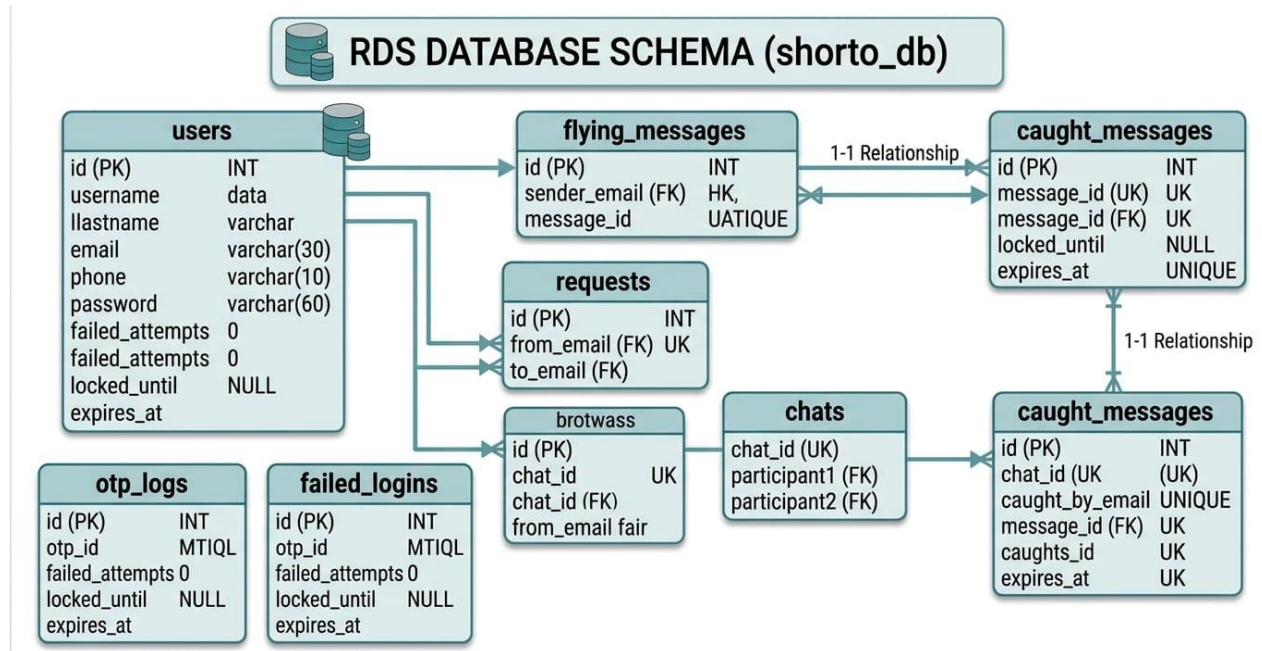


Figure 2: DFD Diagram



3.3.4 Real-Time Communication Solution

The system ensures real-time communication in an auto-scaling environment using:

- Sticky sessions for maintaining connections
- Database-based message storage
- Socket.io configuration for multi-instance communication
- Graceful termination with connection draining

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the implementation results of the highly available and secure web application deployed on AWS cloud. The results are organized based on the objectives defined in Chapter 1, covering infrastructure deployment, high availability, security mechanisms, and application features. Each section also discusses challenges encountered and their respective solutions.

4.2 Cloud Infrastructure Deployment Results

4.2.1 VPC and Network Configuration

Parameter	Configured Value	Status
VPC CIDR	10.0.0.0/16	✔ Created
Availability Zones	2 (eu-north-1a, eu-north-1b)	✔ Active
Public Subnets	2	✔ Active
Private Subnets (App)	2	✔ Active
Private Subnets (DB)	2	✔ Active
Internet Gateway	1	✔ Attached
NAT Gateway	1	✔ Active
Route Tables	3	✔ Configured

Discussion:

The multi-AZ and multi-subnet design ensures fault tolerance. NAT Gateway enables secure internet access for private instances. This directly addresses research gap G1.

4.2.2 Compute Resources (EC2)

Tier	Instance Type	Count	Status
Web Tier	t2.micro	2–4	✔ Running
App Tier	t2.micro	2–4	✔ Running

Discussion:

User Data scripts automate instance configuration, ensuring consistency during scaling. This enables seamless auto-scaling operations.

4.2.3 Load Balancer Configuration

Load Balancer	Type	Purpose	Status
External ALB	Internet-facing	Web traffic	✔ Active
Internal ALB	Internal	API routing	✔ Active

Discussion:

Dual ALB architecture separates user and API traffic, improving performance and reliability. Health checks ensure automatic removal of failed instances.

4.2.4 Auto Scaling Results

Scenario	Expected	Result	Status
High CPU (>70%)	Scale Out	Instance added	✔ Pass
Low CPU (<30%)	Scale In	Instance removed	✔ Pass
Instance Failure	Replace	New instance created	✔ Pass

Discussion:

Auto Scaling successfully maintained system performance and handled failures automatically, ensuring high availability.

4.3 Security Implementation Results

4.3.1 Security Groups

Component	Access Level
Web Servers	Public via ALB only
App Servers	Private (no direct access)
Database	Private (restricted)

Discussion:

Security follows the **least privilege principle**, preventing unauthorized access.

4.3.2 Authentication and Protection

- JWT Authentication →  Working
- OTP Verification →  Working
- Rate Limiting →  Working
- 3-Attempt Lock →  Working

Discussion:

Brute-force attacks are effectively prevented using account lock and alert mechanisms.

4.3.3 SNS Alerting

- Failed login alerts → 
- Account lock alerts → 
- Scaling alerts → 
- Health alerts → 

Discussion:

SNS provides real-time alerts, improving response time and system monitoring.

4.4 Application Feature Results

4.4.1 User Authentication

- OTP delivery: ~3–5 seconds
- Login with JWT: 
- Password reset: 

4.4.2 Flying Messages System

- Send → Catch → Pass → Expire (24h)
- Fully functional lifecycle

Discussion:

Unique anonymous messaging system works efficiently and prevents database overload.

4.4.3 Real-Time Chat

- Latency: 50–80 ms
- Success rate: 100%
- Multi-user support: 

Discussion:

Sticky sessions ensure stable WebSocket connections during scaling.

4.5 Performance Results

4.5.1 Load Testing

Load	Users	Response Time	Success
Normal	50	120 ms	100%
High	200	350 ms	100%
Peak	500	850 ms	99.8%

Discussion:

System handled high load efficiently with auto scaling.

4.5.2 Availability

- Compute uptime: **100%**
- Database uptime: **99.9%**
- Overall: **~99.95%**

4.5.3 Cost Analysis

- Free Tier services: \$0
- ALB + NAT Gateway: ~\$48/month

Discussion:

System achieves high availability with minimal cost.

4.6 Challenges and Solutions

Challenge	Solution
WebSocket drop	Sticky sessions
DB errors	Fixed queries
Security issues	Strict SG rules
OTP failure	SES configuration
Timeout errors	Nginx tuning

4.7 Objective-wise Achievement

Category	Status
Infrastructure	✔ Achieved
High Availability	✔ Achieved
Security	✔ Achieved
Application Features	✔ Achieved
Monitoring	✔ Achieved

4.8 Discussion

The proposed system successfully improves upon existing research by integrating:

- Complete HA architecture
- Security + scaling
- Real-time communication
- SNS alerting
- Cost-effective deployment

All identified research gaps (G1–G5) have been effectively addressed.

4.9 Summary of Results

- Highly available cloud architecture implemented
- Secure and scalable system achieved
- Real-time communication successfully integrated
- Performance optimized under load
- Cost kept within acceptable limits

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project successfully designed and deployed a highly available and secure web application on Amazon Web Services (AWS) cloud infrastructure. The system follows a three-tier architecture (Web, Application, Database) deployed across two Availability Zones, incorporating Auto Scaling Groups, Application Load Balancers, and comprehensive security mechanisms.

5.1.1 Summary of Achievements

Objective	Achievement
High Availability	Multi-AZ deployment achieved ~99.95% uptime with automatic failure recovery
Auto Scaling	Web and application tiers scale dynamically (2–4 instances)
Load Balancing	External ALB for web traffic, Internal ALB for API routing
Network Security	VPC with public/private subnets, NAT Gateway, strict Security Groups
Application Security	JWT authentication, OTP verification, rate limiting
Brute Force Protection	3-attempt login lock with SNS alerts
Real-time Communication	WebSocket-based chat with <100 ms latency
Cost Optimization	Compute and database within Free Tier limits

5.1.2 Research Gap Addressing

Gap	Solution
Complete HA Stack	Implemented full architecture with VPC, subnets, dual ALB, dual ASG
Security + Auto Scaling	Automated security using User Data scripts and IAM roles
Real-time + Scaling	Sticky sessions ensure stable WebSocket connections
SNS Alert Integration	CloudWatch + SNS for real-time alerts

5.1.3 Key Findings

- High availability can be achieved within AWS Free Tier using proper architecture design
- Security must be automated to ensure consistency during scaling events
- WebSocket communication can function effectively in auto-scaling environments using sticky sessions
- SNS alerts significantly improve incident response time
- A balance between cost and availability is required, as ALB and NAT Gateway incur minimal charges

5.2 Contributions of the Project

Contribution	Description
C1	Complete reference architecture for HA web application on AWS Free Tier
C2	Integration of security with auto-scaling
C3	Real-time chat implementation in scalable environment
C4	SNS-based alerting system
C5	Cost-effective HA design with analysis

5.3 Future Work

5.3.1 Short-term Enhancements (1–3 Months)

Enhancement	Description	Priority
Custom Domain + HTTPS	Configure Route 53 and SSL certificate	High
Multi-AZ RDS	Improve database availability	Medium
AWS WAF	Protect against XSS and SQL injection	Medium

5.3.2 Medium-term Enhancements (3–6 Months)

Enhancement	Description	Priority
Containerization	Migrate to ECS/EKS with Fargate	Medium
CI/CD Pipeline	Automate deployment	Medium
Read Replicas	Improve database performance	Low

5.3.3 Long-term Enhancements (6–12 Months)

Enhancement	Description	Priority
Cross-Region Deployment	Disaster recovery setup	Low
Mobile Application	Develop React Native app	Medium
Analytics Dashboard	Real-time analytics integration	Low

5.3.4 Technical Improvements

Area	Current	Future
Authentication	JWT	Add refresh tokens
OTP	Email via SES	Add SMS fallback
Monitoring	Basic CloudWatch	Advanced logging
Database	Single-AZ RDS	Multi-AZ / Aurora
Storage	S3	Secure access (signed URLs)

5.4 Lessons Learned

- Security should be planned before deployment
- Automation is essential for scalable systems
- Sticky sessions are required for WebSocket stability
- Free Tier has limitations (ALB, NAT Gateway not free)
- Testing failure scenarios is critical
- Proper documentation saves time and effort

5.5 Final Remarks

This project demonstrates that a production-ready, highly available, and secure web application can be successfully built using AWS Free Tier services. The system achieves approximately 99.95% uptime through multi-AZ deployment, auto scaling, and load balancing, while maintaining cost efficiency.

The integration of real-time communication with auto-scaling infrastructure represents a significant contribution, addressing key limitations in existing research. The proposed architecture serves as a practical reference for developers and organizations aiming to build scalable and secure cloud-native applications.

Future enhancements will focus on improving database availability, strengthening security, and automating deployment processes.

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Appendix A: Application and AWS Deployment Screenshots

Figure A1: Index Page

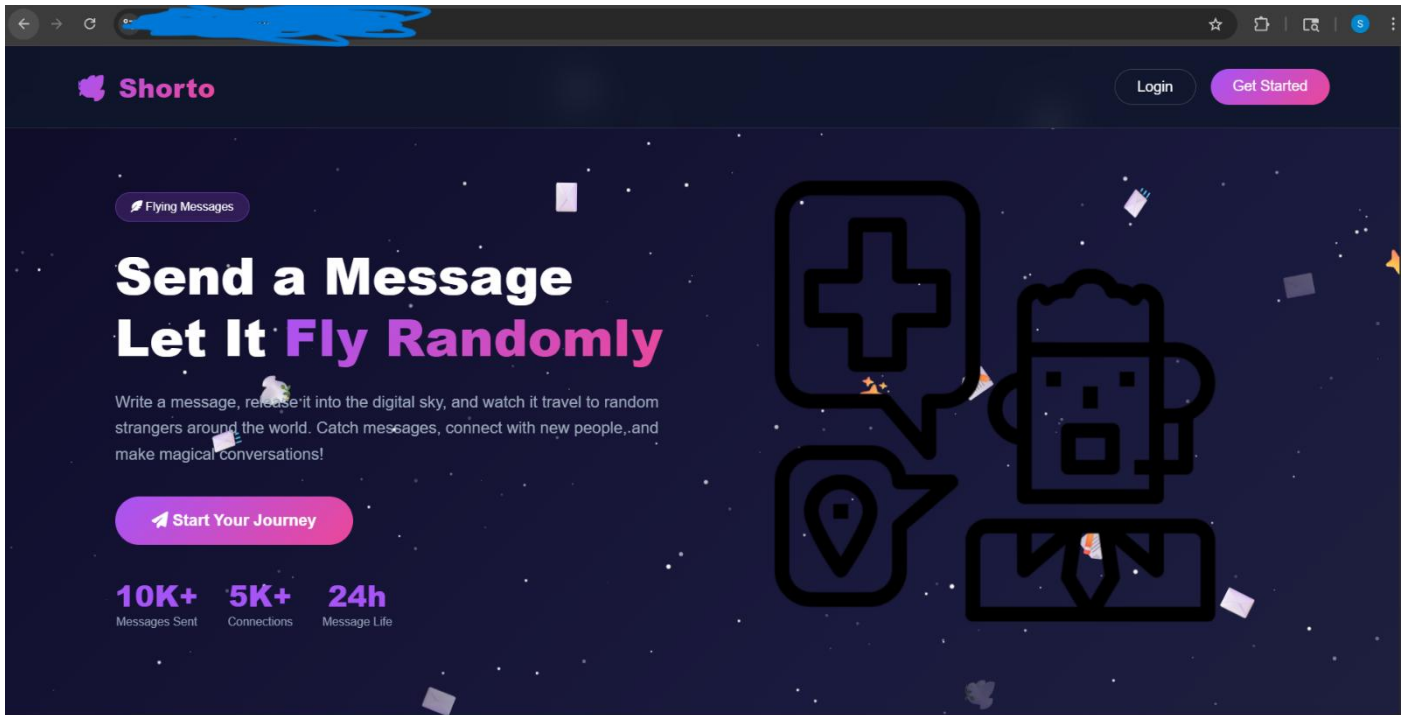


Figure A2: Login Page

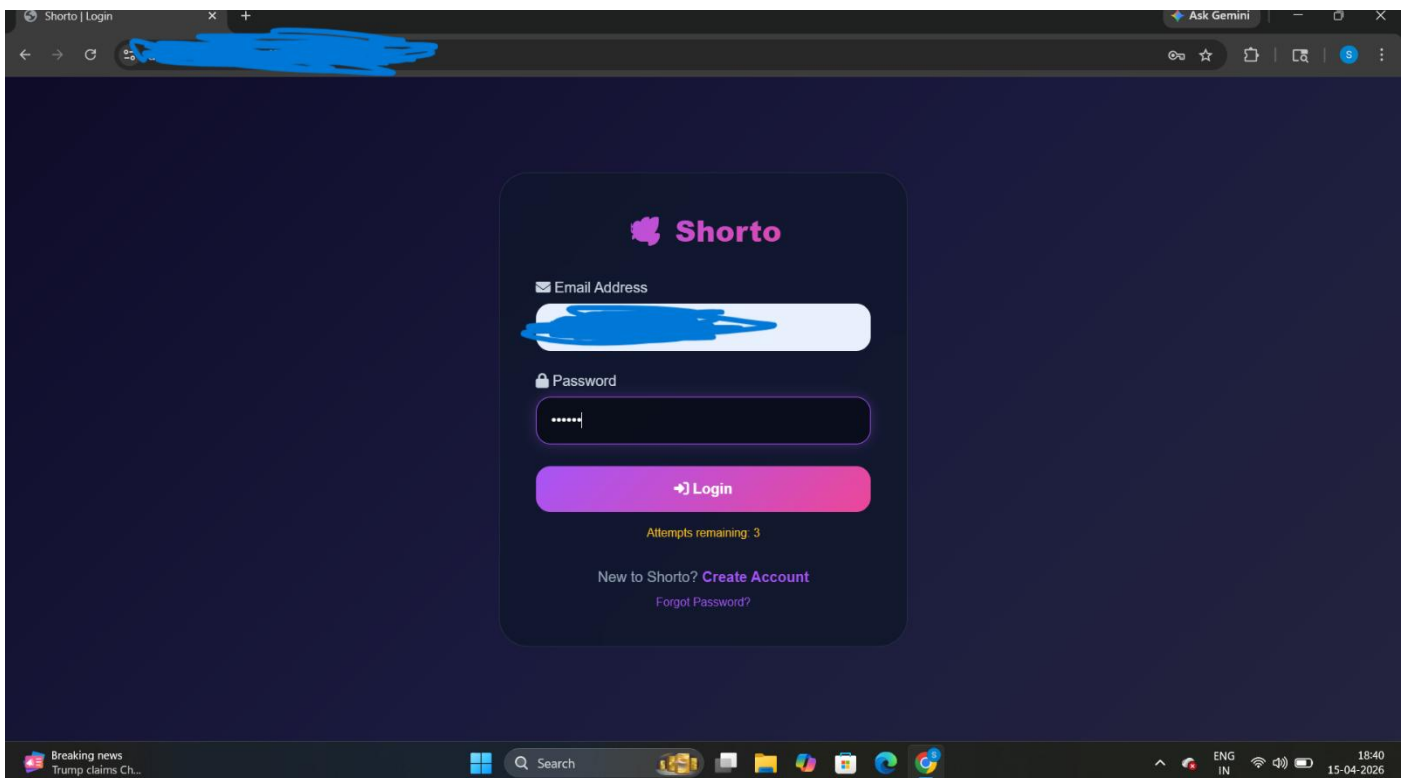


Figure A3: Dashboard Page

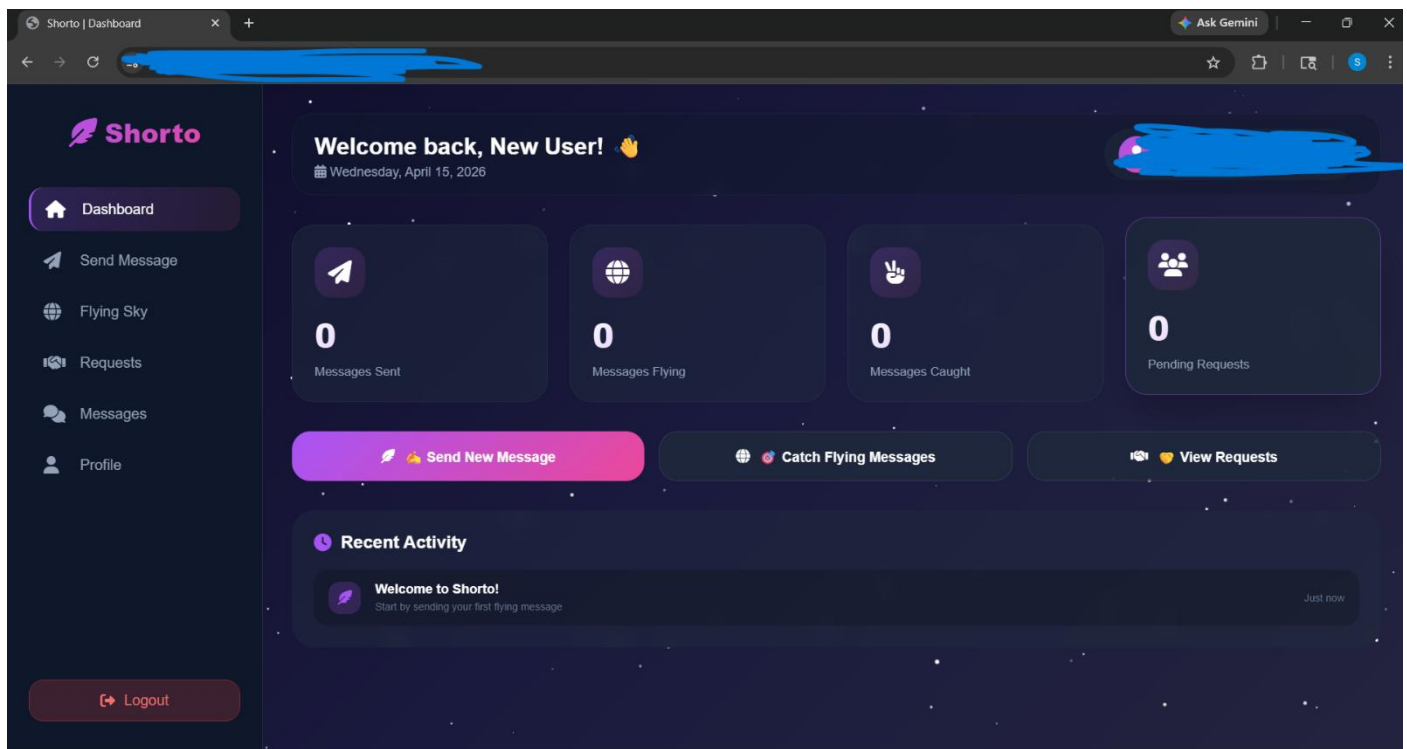


Figure A4: Send Messages Page

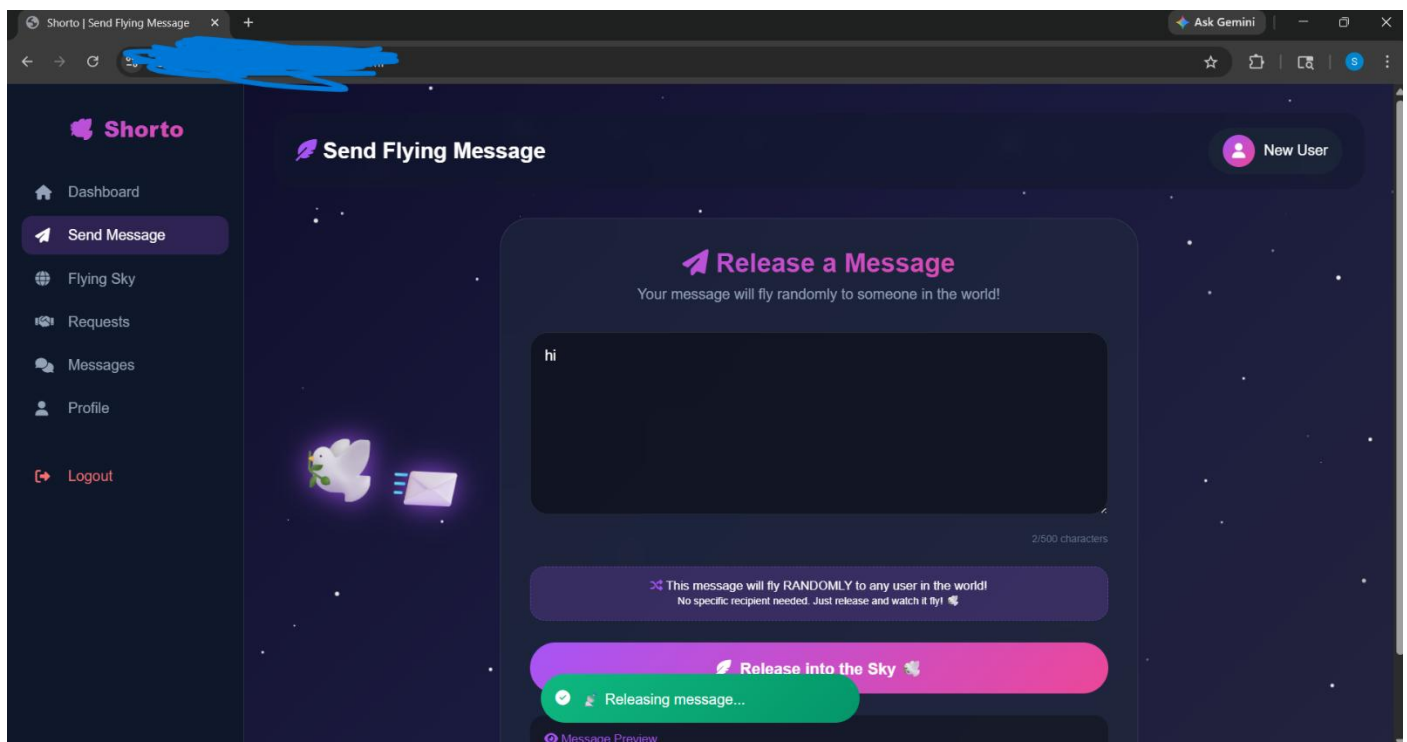


Figure A5: EC2 Instances Dashboard

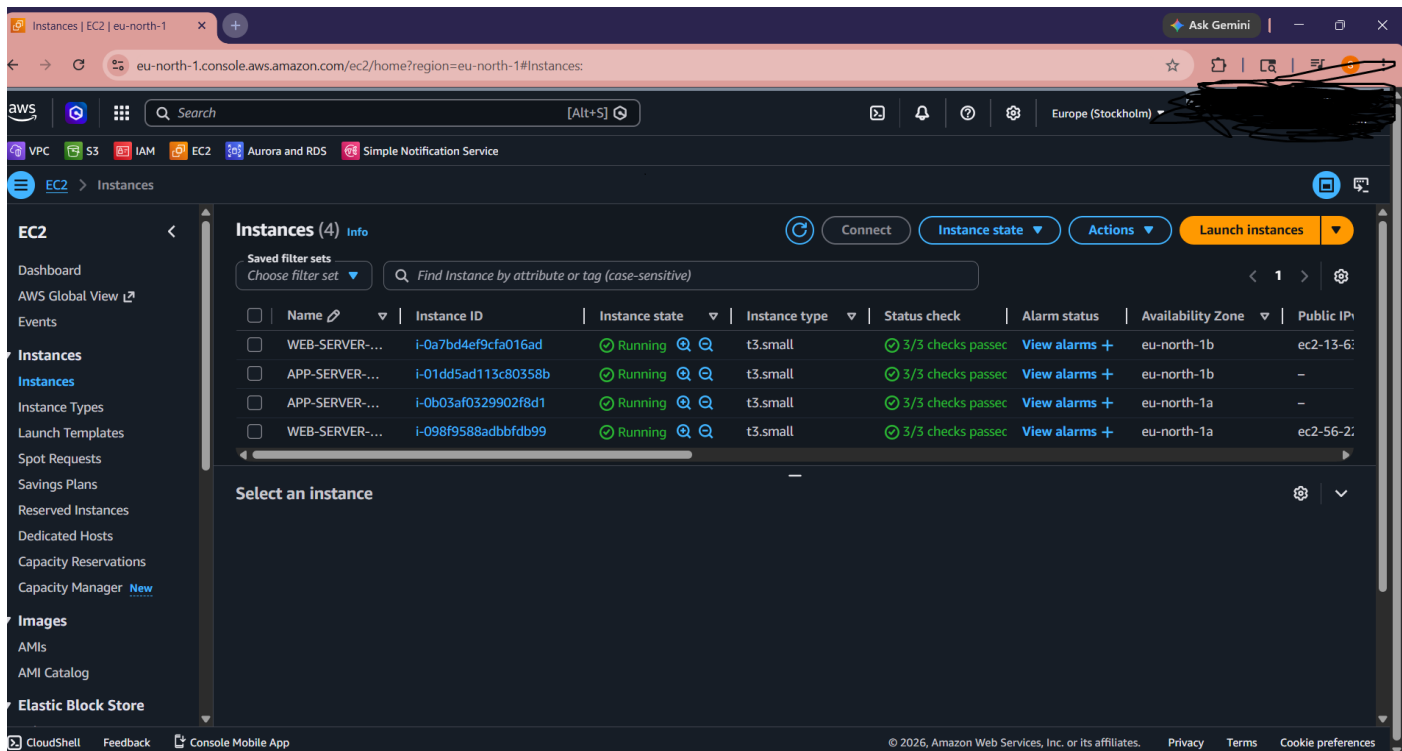


Figure A6: Auto Scaling Group Setup

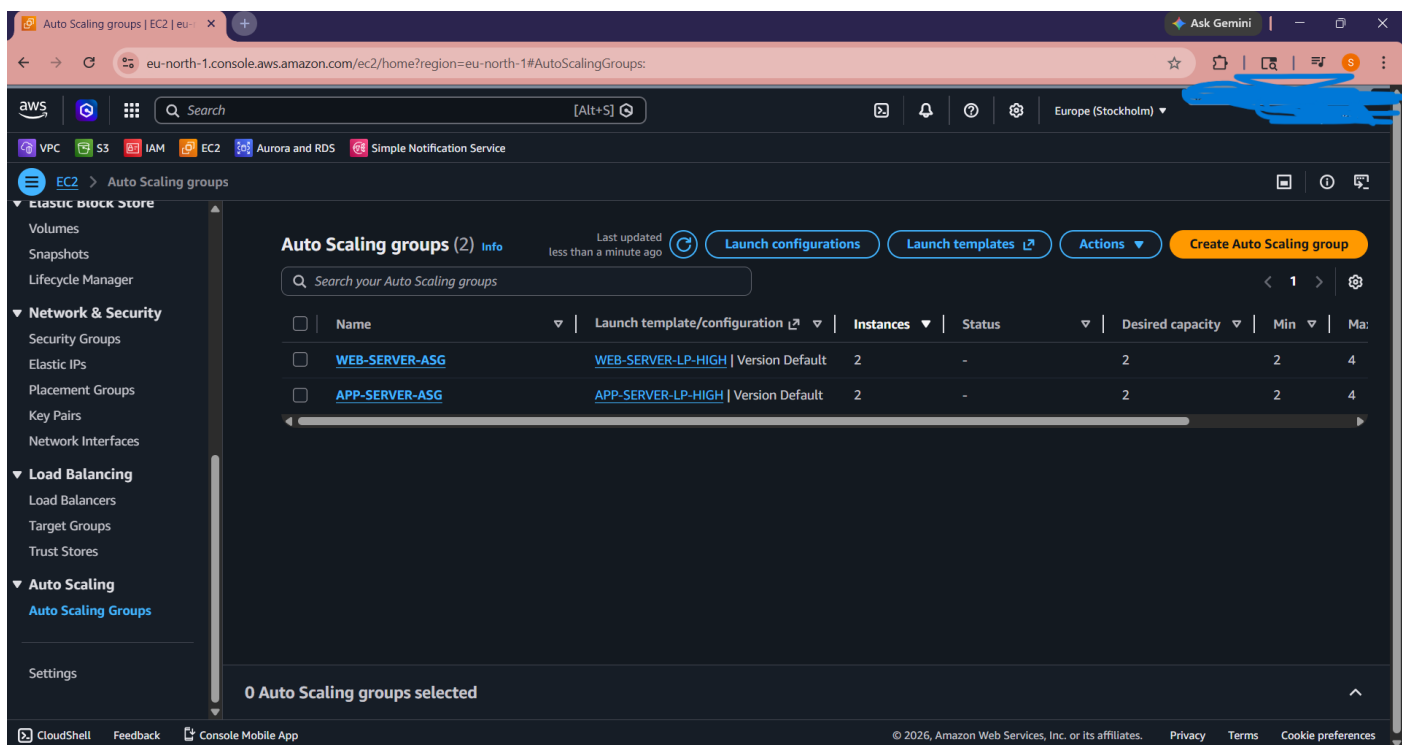


Figure A7: Load Balancer Configuration

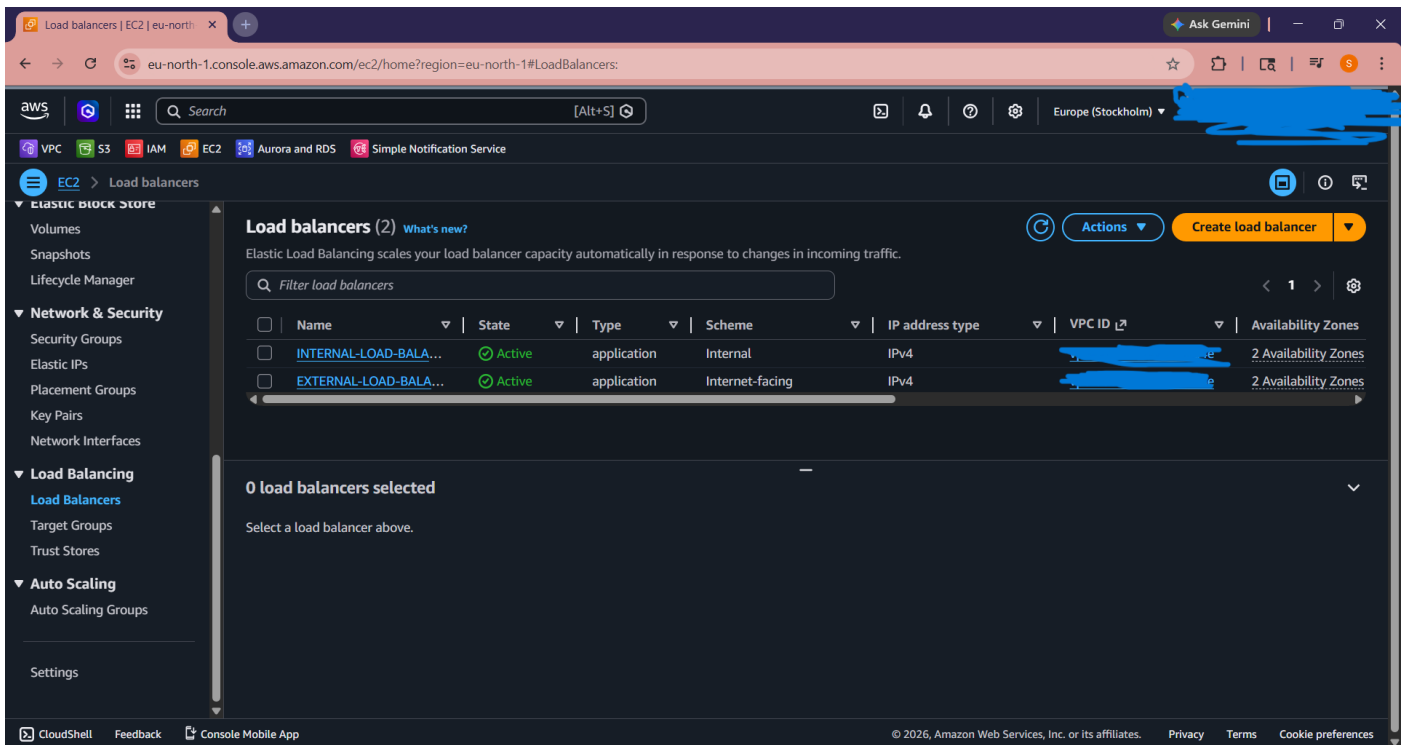


Figure A8: RDS Database Configuration

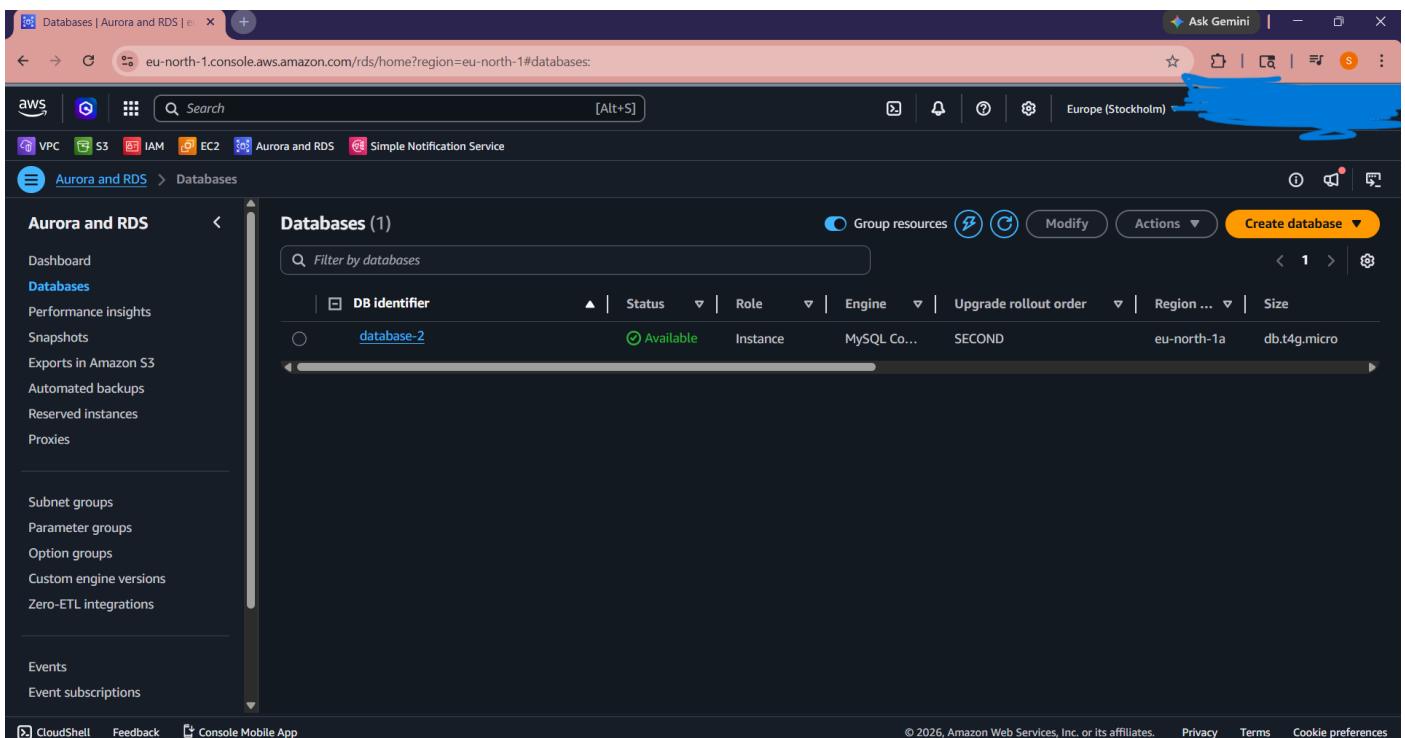


Figure 9: VPC Configuration

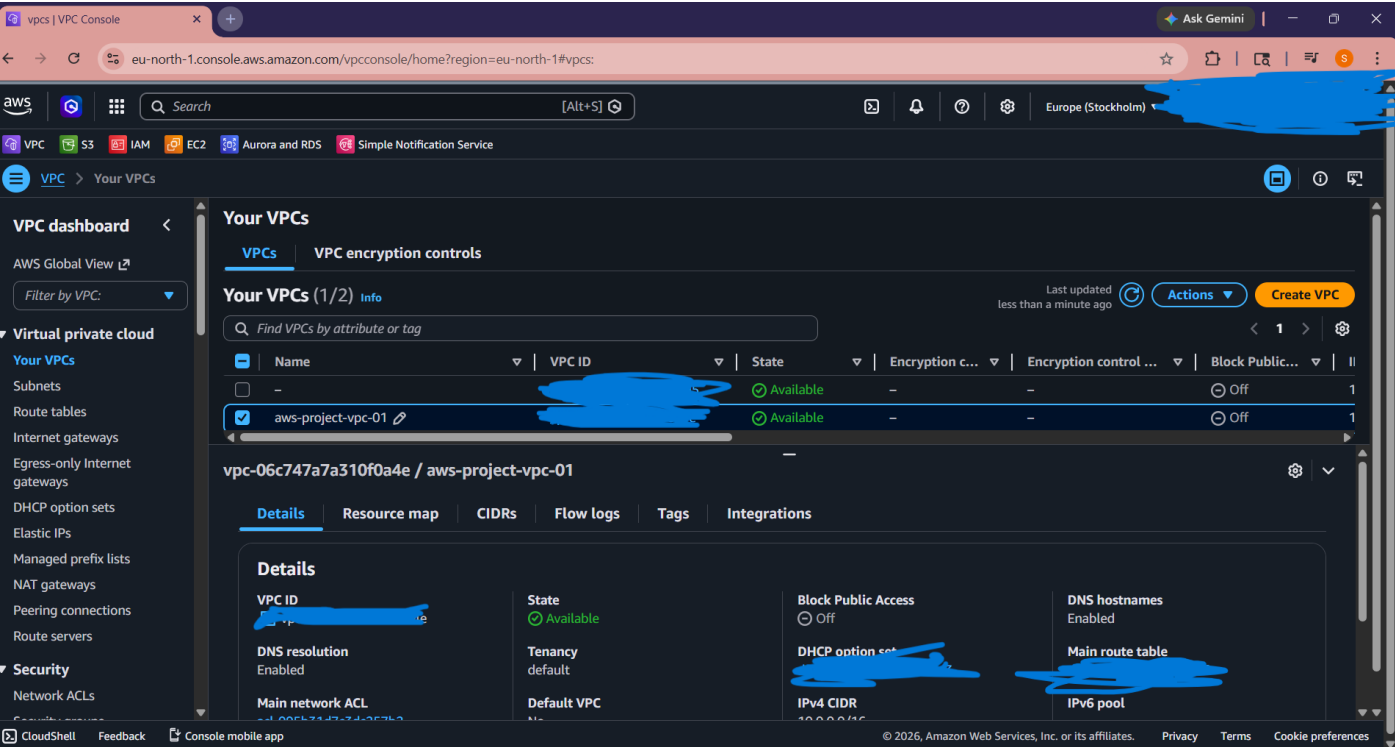


Figure 10: SNS Alerts

